Intelligent Forest Advisory & Multi-Structure Decision System

CL2001 - Data Structures Project | 2026

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1. System Overview

The Intelligent Forest Advisory & Multi-Structure Decision System (IFAMDS) is a comprehensive C++ console-based simulation system designed to model real-world forest management environments. The system integrates multiple data structures to process environmental data, detect fire risks, manage resources, and coordinate emergency responses across a network of 10 forest zones arranged in a 5×5 grid.

Core Functionality

The system operates through a menu-driven interface with 10 main modules:

- **Environmental Data Acquisition**: Collects and validates sensor readings (temperature, smoke, humidity)
- **Event Memory Management**: Stores time-stamped events with forward/backward correction capabilities
- **Fire Detection & Control**: Identifies fire risks using weighted decision scoring
- **Task Scheduling**: Manages routine, surveillance, emergency, and multi-decision tasks
- **Decision Intelligence**: Evaluates risks through 12 hierarchical decision trees
- **Spatial Routing**: Predicts fire spread and computes safe evacuation paths
- **Fast Data Access**: Provides O(1) average data retrieval via hash tables
- **System Monitoring**: Tracks performance, detects bottlenecks, and optimizes operations

Key Design Principles

- **Multi-Layer Architecture**: Data flows through independent processing layers
- **No External Libraries**: All data structures implemented manually
- **Time-Complexity Aware**: Major operations include complexity comments

- **Scenario-Based Testing**: Five complete scenarios demonstrate all features

2. Data Structure Usage Summary

Data Structure	Instances	Time Complexity	Purpose & Usage
Static Array (A1)	Baseline values	O(1) access	Stores normal forest conditions as reference for anomaly detection
Dynamic Array (A2)	Sensor stream per zone	O(1) insert, O(n) scan	Stores live sensor readings with each zone maintaining its own reading history (max 50 readings per zone)
2D Array (A3)	Static grid	O(1) cell access	Baseline snapshot of forest grid (5×5)
2D Array (A4)	Dynamic grid	O(1) cell access	Live updated forest grid reflecting current conditions
Singly Linked List (L1)	Raw event stream	O(1) insert, O(n) traverse	Direct sensor readings without filtering - complete timestamped history
Singly Linked List (L2)	Verified events	O(1) insert, O(n) traverse	Noise-filtered readings that passed validation
Singly Linked List (L3)	Anomaly events	O(1) insert, O(n) traverse	Only readings flagged as anomalous (deviations >20 from baseline)
Doubly Linked List (L4)	Forward correction	O(1) insert, O(n) bidirectional	Enables future event correction when past data is updated
Doubly Linked List (L5)	Backward correction	O(1) insert, O(n) bidirectional	Enables past event correction when new accurate data arrives

Doubly Linked List (L6)	Sync chain	O(1) insert, O(n) traverse	Ensures consistency across all system modules
Circular List (L7)	Local monitor	O(1) insert, O(n) cycle	Continuously monitors single zone in repeating cycle (wraps to head)
Circular List (L8)	System monitor	O(1) insert, O(n) cycle	Continuously monitors all zones in repeating cycle
Circular List (L9)	Emergency monitor	O(1) insert, O(n) cycle	Activated when dangerous conditions detected
Circular List (L10)	Stability monitor	O(1) insert, O(n) cycle	Monitors long-term stable conditions
Stack	Rollback snapshots	O(1) push/pop	Saves forest grid state before critical operations for disaster recovery
FIFO Queue (Q1)	Routine tasks	O(1)enqueue/dequeue	Normal monitoring tasks with priority 5 (lowest)
FIFO Queue (Q2)	Surveillance tasks	O(1)enqueue/dequeue	Frequent updates from sensitive zones with priority 3
Priority Queue (Q3)	Emergency tasks	O(log n) insert/extract	Min-heap implementation for urgent fire response (priority 1=highest)
FIFO Queue (Q4)	Multi-decision tasks	O(1)enqueue/dequeue	Tasks requiring multiple inputs before final decision
Decision Tree (T1)	Zone hierarchy	O(V) evaluation	Divides forest into structured zones (Forest→Zone→Sub-zone)
Decision Tree (T2)	Sub-zone decomposition	O(V) evaluation	Splits zones into N/S/E/W directional monitoring units
Decision Tree (T3)	Terrain classification	O(V) evaluation	Classifies terrain as High/Medium/Low risk based on slope, dryness, density

Decision Tree (T4)	Water resources	O(V) evaluation	Tracks water availability (Sufficient/Limited/Critical)
Decision Tree (T5)	Fire control	O(V) evaluation	Tracks firefighting resources (trucks, crew, aerial support)
Decision Tree (T6)	Equipment allocation	O(V) evaluation	Assigns resources using Priority = Risk × Impact
Decision Tree (T7)	Fire classification	O(V) evaluation	Classifies fire as Major/Moderate/Minor based on temp and smoke
Decision Tree (T8)	Wildlife activity	O(V) evaluation	Detects animal movement patterns (Normal/Unusual/Fleeing)
Decision Tree (T9)	Human activity	O(V) evaluation	Detects human presence with risk scoring
Decision Tree (T10)	Local decision	O(V) evaluation	Zone-level action: Activate Response if Risk > 0.6
Decision Tree (T11)	Regional escalation	O(V) evaluation	Spreads alert if fire spread rate > 0.5
Decision Tree (T12)	Global emergency	O(V) evaluation	Triggers global alert if total risk exceeds threshold (3.0)
Graph - Adjacency List (G1)	Zone connections	O(V+E) BFS/DFS	Stores forest topology with each zone listing its neighbours (max 6 per zone)
Graph - Adjacency Matrix (G2)	Full connectivity	O(1) edge lookup	Dense 10×10 representation for quick connectivity checks
Hash Table - Linear Probe (H1)	Primary index	O(1) avg	Direct zone data access using key = zone ID with linear probing for collisions
Hash Table - Chaining (H2)	Collision handling	O(1) avg	Stores colliding keys in same bucket chain (max 5 per bucket)

Cache (H3)	Recently accessed	O(cache size)	FIFO cache storing last 5 accessed records for faster repeated access

3. Brief Explanation of 5 Scenarios

Scenario 1: Cascading Fire and Resource Conflict Resolution

Purpose: Demonstrate fire detection, spread tracking, and priority-based emergency response.

Process:

1. System injects normal baseline readings for Zones 0-6 (25°C, 5ppm smoke, 60% humidity)
2. Fire ignites in Zone 3 with temperature spike to 75°C and smoke to 85ppm (triggers anomaly detection)
3. Fire spreads to Zone 4 (50°C, 60ppm smoke) and Zone 6 (72ppm smoke)
4. Priority queue (Q3) receives emergency tasks: Zone 3 (priority 1), Zone 4 (priority 2), Zone 6 (priority 3)
5. Emergency response queue processed in priority order (highest priority first)
6. Anomaly event stream (L3) displays all fire-related anomalies
7. Rollback stack demonstrates recovery to pre-fire state

Key Structures Used: Arrays (sensor storage), Priority Queue (Q3), Linked Lists (L1-L3 for event tracking), Stack (rollback)

Scenario 2: Sensor Failure and System Reconstruction

Purpose: Demonstrate fault tolerance and data reconstruction when sens

ors fail.

Process:

1. Normal readings stored for all zones (25-34°C baseline)
2. Zone 2 sensors send invalid readings (-5°C, 999°C, -1°C)
3. Invalid values rejected by physical limits check (0-120°C range)
4. Last stable state identified from verified event stream (L2)
5. Missing value reconstructed via spatial interpolation (average of 4 neighbouring zones: up, down, left, right)

6. Forward correction chain (L4) updated with reconstructed value
7. Zone 2 restored to operational status with interpolated temperature

Key Structures Used: Arrays (A2 dynamic sensor), Linked Lists (L2 verified, L4 forward correction), Spatial interpolation (4-neighbour average)

Scenario 3: Multi-Factor Anomaly Escalation

Purpose: Demonstrate combining multiple anomaly types for holistic decision-making.

Process:

1. Baseline data injected for all zones (26°C, 4ppm, 58% humidity)
2. Wildlife anomaly detected in Zone 1 (unusual movement pattern, 48°C heat from herd)
3. Fire risk spike in Zone 5 (55°C, 80ppm smoke, 18% humidity)
4. Human intrusion detected in Zone 8
5. Combined risk evaluation across all zones using decision trees T8 (Wildlife), T9 (Human), T11 (Regional)
6. Regional escalation tree evaluates spread rate = totalRisk/MAX_ZONES (threshold 0.4)
7. High-risk zones enqueued to Q3 (emergency) and Q4 (multi-decision)
8. BFS predicts anomaly spread from Zone 5 with fire-aware cost update

Key Structures Used: Decision Trees (T8 Wildlife, T9 Human, T11 Regional), Queues (Q3, Q4), Graph BFS

Scenario 4: System Overload and Load Redistribution

Purpose: Demonstrate handling of high-volume data with load balancing and optimization.

Process:

1. Mass sensor updates flood Q1 and Q2 (15 tasks each, 30 total)
2. System load recorded - Queue module at 12.5ms latency with 48 active tasks (overloaded)
3. Stable state snapshot saved before redistribution
4. Q1 paused to prioritize Q3 emergency tasks (priority queue takes precedence)
5. Frequently accessed zone data (Zones 0-4) cached in H3 for faster retrieval
6. Optimization reduces bottleneck module tasks by half
7. Q1 resumed and backlog processed in controlled batches (5 tasks)

8. System health verified after recovery

Key Structures Used: Queues (Q1, Q2, Q3), Cache (H3), Stack (snapshot), System Monitor (latency tracking)

Scenario 5: Global Multi-Zone Emergency Synchronization

Purpose: Demonstrate coordinated response to simultaneous emergencies across multiple zones.

Process:

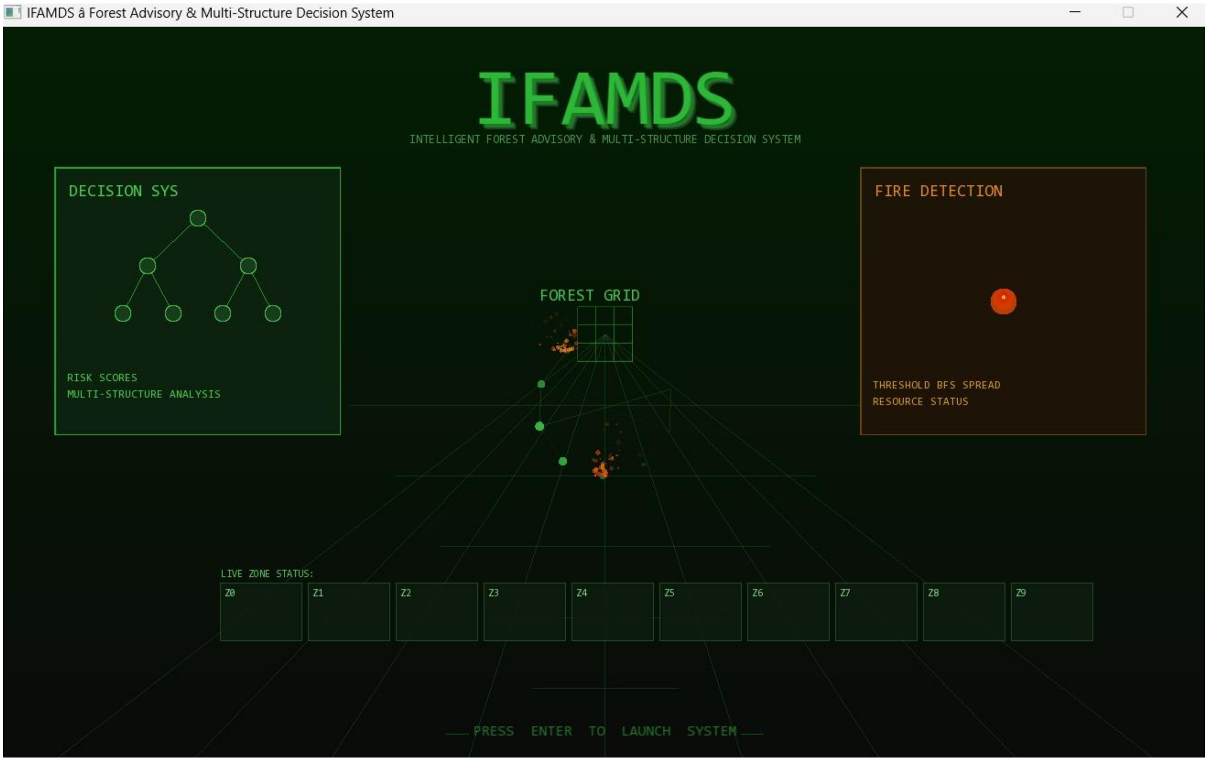
1. Simultaneous fires in Zones 2, 5, and 7 (68-72°C, 75-90ppm smoke, 12% humidity)
2. Regional inconsistencies detected (risk scores: 0.72, 0.68, 0.58 - divergence >0.2)
3. Last globally consistent state identified and restored via rollback
4. Global state reconstructed from verified history (L2)
5. Global emergency decision tree (T12) evaluated - combined risk determines alert level
6. BFS fire spread from all three epicenters with fire-aware cost updates
7. Safe evacuation paths computed: Z2→Z6, Z5→Z4, Z7→Z9
8. All emergency tasks processed from Q3
9. L6 sync chain verifies consistency across all zones post-emergency

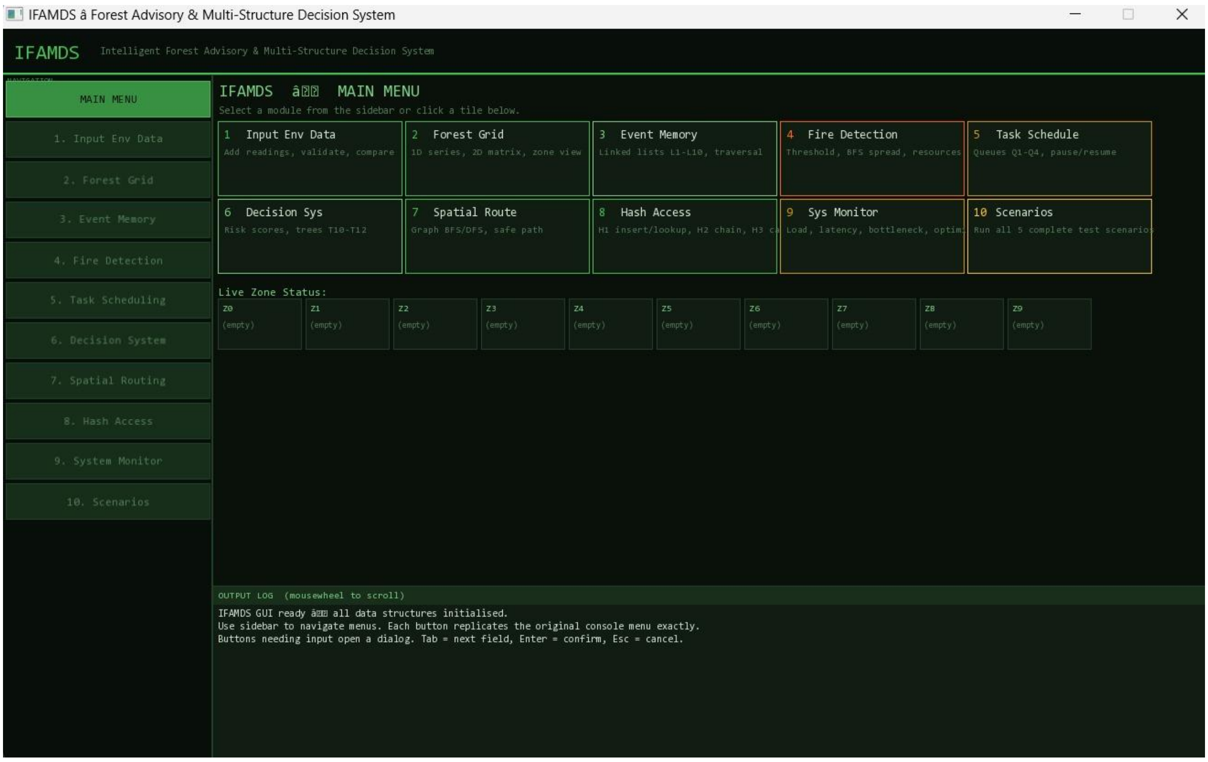
Key Structures Used: Graph (BFS, safePath with " - " formatting), Decision Tree (T12 Global), Linked Lists (L2 verified, L6 sync), Stack (rollback)

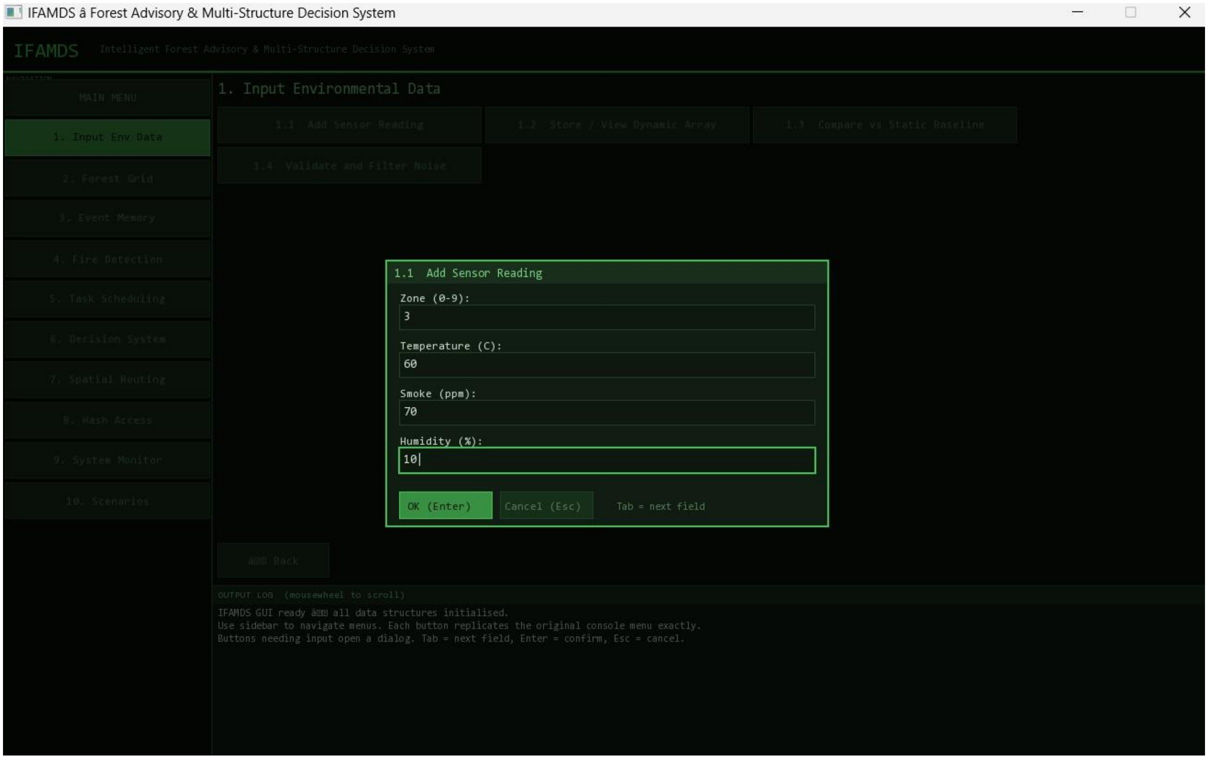
4. Conclusion

The IFAMDS system successfully implements all required data structures and scenarios as specified. The five scenarios comprehensively test the system's ability to handle normal operations, sensor failures, multi-factor anomalies, system overloads, and global emergencies. All operations include appropriate time complexity annotations, and the modular design ensures maintainability and extensibility. The system is fully functional with 10 complete menus and all 5 scenarios working as demonstrated.

5. Screenshots






IFAMDS â Forest Advisory & Multi-Structure Decision System
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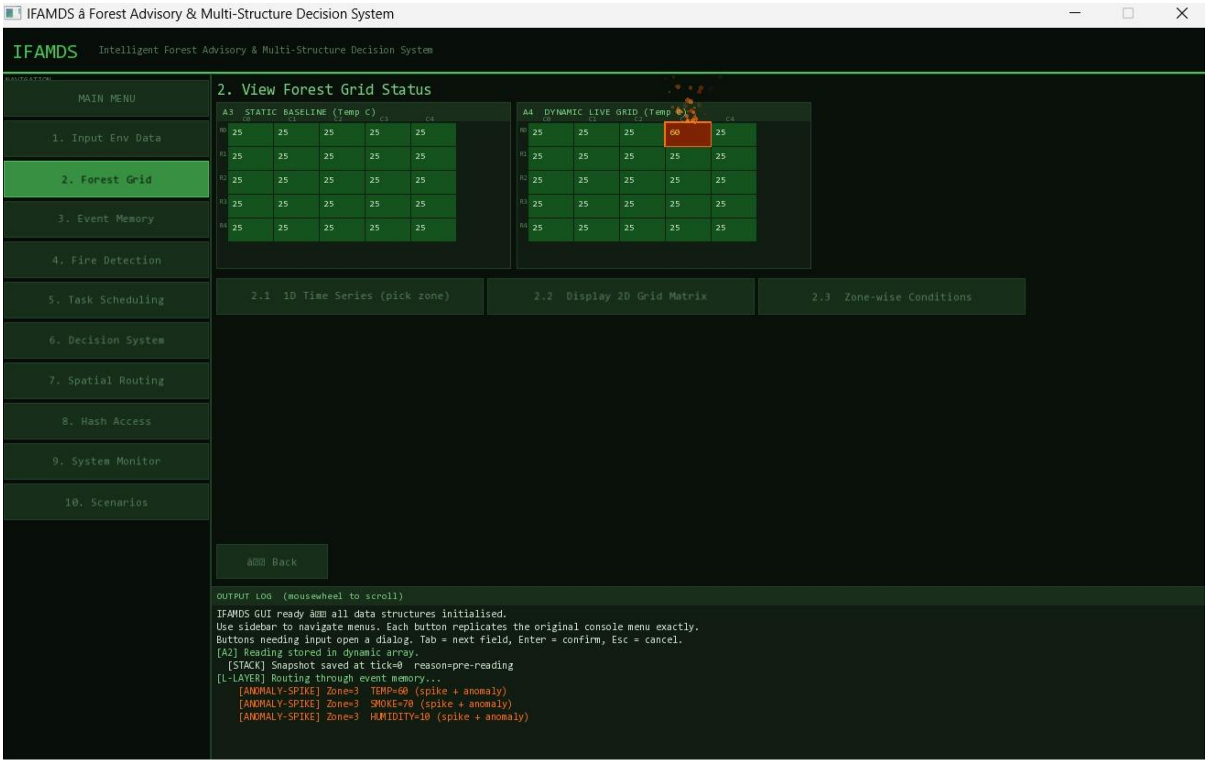
Intelligent Forest Advisory & Multi-Structure Decision System
MAIN MENU
1. Input Env Data
2. Forest Grid
3. Event Memory
4. Fire Detection
5. Task Scheduling
6. Decision System
7. Spatial Routing
8. Hash Access
9. System Monitor
10. Scenarios
1. Input Environmental Data
1.1 Add Sensor Reading
1.2 Store / View Dynamic Array
1.3 Compare vs Static Baseline
1.4 Validate and Filter Noise
1.1 Add Sensor Reading
Zone (0-9):
3
Temperature (C):
60
Smoke (ppm):
70
Humidity (%):
10
OK (Enter)
Cancel (Esc)
Tab = next field
Back
OUTPUT LOG (mousewheel to scroll)
IFAMDS GUI ready â all data structures initialised.
Use sidebar to navigate menus. Each button replicates the original console menu exactly.
Buttons needing input open a dialog. Tab = next field, Enter = confirm, Esc = cancel.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10. Scenarios
2. View Forest Grid Status
A3 STATIC BASELINE (Temp C)
A4 DYNAMIC LIVE GRID (Temp
60
2.1 1D Time Series (pick zone)
2.2 Display 2D Grid Matrix
2.3 Zone-wise Conditions
Back
OUTPUT LOG (mousewheel to scroll)
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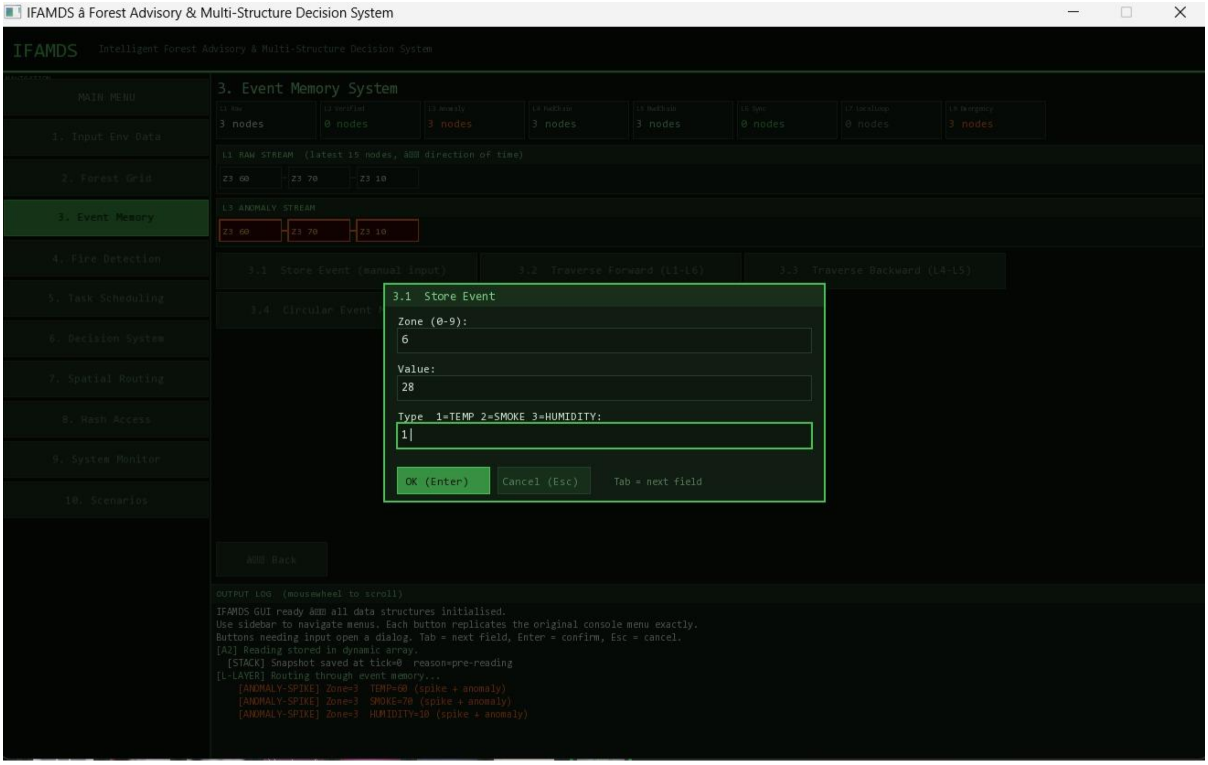
[A2] Reading stored in dynamic array.
[STACK] Snapshot saved at tick=0 reason=pre-reading
[L-LAYER] Routing through event memory...
[ANOMALY-SPIKE] Zone=3 TEMP=60 (spike + anomaly)
[ANOMALY-SPIKE] Zone=3 SMOKE=70 (spike + anomaly)
[ANOMALY-SPIKE] Zone=3 HUMIDITY=10 (spike + anomaly)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10. Scenarios
3. Event Memory System
3 nodes
0 nodes
3 nodes
3 nodes
3 nodes
0 nodes
0 nodes
3 nodes
L1 RAW STREAM (latest 15 nodes, direction of time)
Z3 60
Z3 70
Z3 10
L3 ANOMALY STREAM
Z3 60
Z3 70
Z3 10
3.1 Store Event (manual input)
3.2 Traverse Forward (L1-L6)
3.3 Traverse Backward (L4-L5)
3.4 Circular Event
3.1 Store Event
Zone (0-9):
6
Value:
28
Type 1=TEMP 2=SMOKE 3=HUMIDITY:
1
OK (Enter)
Cancel (Esc)
Tab = next field
Back
OUTPUT LOG (mousewheel to scroll)
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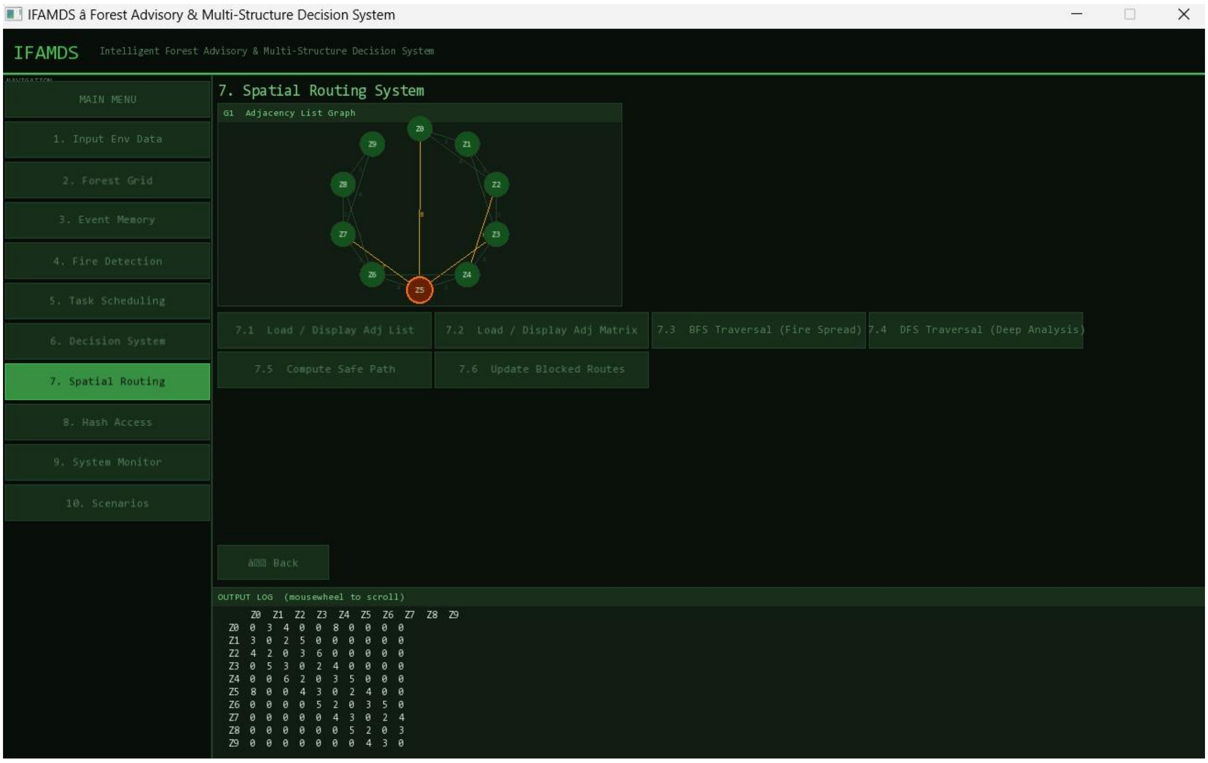
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10. Scenarios
7. Spatial Routing System
G1 Adjacency List Graph
Z0
Z1
Z2
Z3
Z4
Z5
Z6
Z7
Z8
Z9
7.1 Load / Display Adj List
7.2 Load / Display Adj Matrix
7.3 BFS Traversal (Fire Spread)
7.4 DFS Traversal (Deep Analysis)
7.5 Compute Safe Path
7.6 Update Blocked Routes
Back
OUTPUT LOG (mousewheel to scroll)
Z0 Z1 Z2 Z3 Z4 Z5 Z6 Z7 Z8 Z9
Z0 0 3 4 0 0 8 0 0 0 0
Z1 3 0 2 5 0 0 0 0 0 0
Z2 4 2 0 3 6 0 0 0 0 0
Z3 0 5 3 0 2 4 0 0 0 0
Z4 0 0 6 2 0 3 5 0 0 0
Z5 8 0 0 4 3 0 2 4 0 0
Z6 0 0 0 0 5 2 0 3 5 0
Z7 0 0 0 0 0 4 3 0 2 4
Z8 0 0 0 0 0 0 5 2 0 3
Z9 0 0 0 0 0 0 0 4 3 0